BT-201 (GS)

B.Tech., I & II Semester Examination, December 2025

Grading System (GS)

ENGINEERING PHYSICS

Time : Three Hours Maximum Marks : 70

Note : (i) Attempt any five questions. (ii) All questions carry equal marks. (iii) Original paper bilingual (Hindi + English) tha; yahan sirf English version diya gaya hai.

Coverage: Full paper (Q1–Q8)

1. (a) Explain the wave nature of particles. Derive the time-independent Schrödinger equation for a particle in one-dimensional potential box and write its wavefunction. **7**

(b) A particle of mass 9.11×10^{-31} kg is moving in a one-dimensional box of width 10 nm. Calculate the energy of the particle in the first three excited states. **7**

2. (a) Describe how Newton's ring experiment can be used to determine the refractive index of a liquid. **7**

(b) Explain briefly the Rayleigh criterion of resolution. Discuss the resolving power of plane transmission grating and find the relation between resolving power and dispersive power of the grating. **7**

3. (a) Explain Hall effect and derive expression for Hall voltage. Discuss its applications. **7**

(b) Explain Fermi level for intrinsic and extrinsic semiconductors. Discuss the density of states and Bloch's theorem. **7**

4. (a) Derive the relation between Einstein's coefficients (A & B) and explain the concept of population inversion. **7**

(b) Explain the construction and working of He–Ne laser with suitable diagram. **7**

5. (a) Prove the Poynting theorem in electrodynamics and explain the physical significance of each of the terms appearing in the final expression of the theorem. **7**

(b) Derive Maxwell's equations in vacuum. **7**

6. (a) Discuss the uncertainty principle. Derive the relation between group velocity and phase velocity for a free particle. **7**

(b) Calculate the NA, acceptance angle, critical angle and acceptance cone for a fibre having RI of 1.55 and 1.50 respectively for its core & cladding. **7**

7. (a) Explain acceptance angle and acceptance cone of optical fibre. Derive expression for them. **7**

(b) Discuss the phenomenon of Fraunhofer diffraction at a single slit. Show that the intensity of first subsidiary maximum is about 4.5% of the principal maximum. **7**

8. (a) Newton's rings are observed normally incident in reflected light of wavelength 6000 Å. The diameter of the 10th dark ring is 0.50 cm; find the radius of curvature of the lens and thickness of the film. **7**

(b) Explain Michelson interferometer and derive expression for fringe shift due to displacement of mirror. **7**

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